AI RESEARCH LOG · AUG – OCT 2026

# Can a model keep learning after training, without forgetting?

Today's LLMs learn once, then freeze:

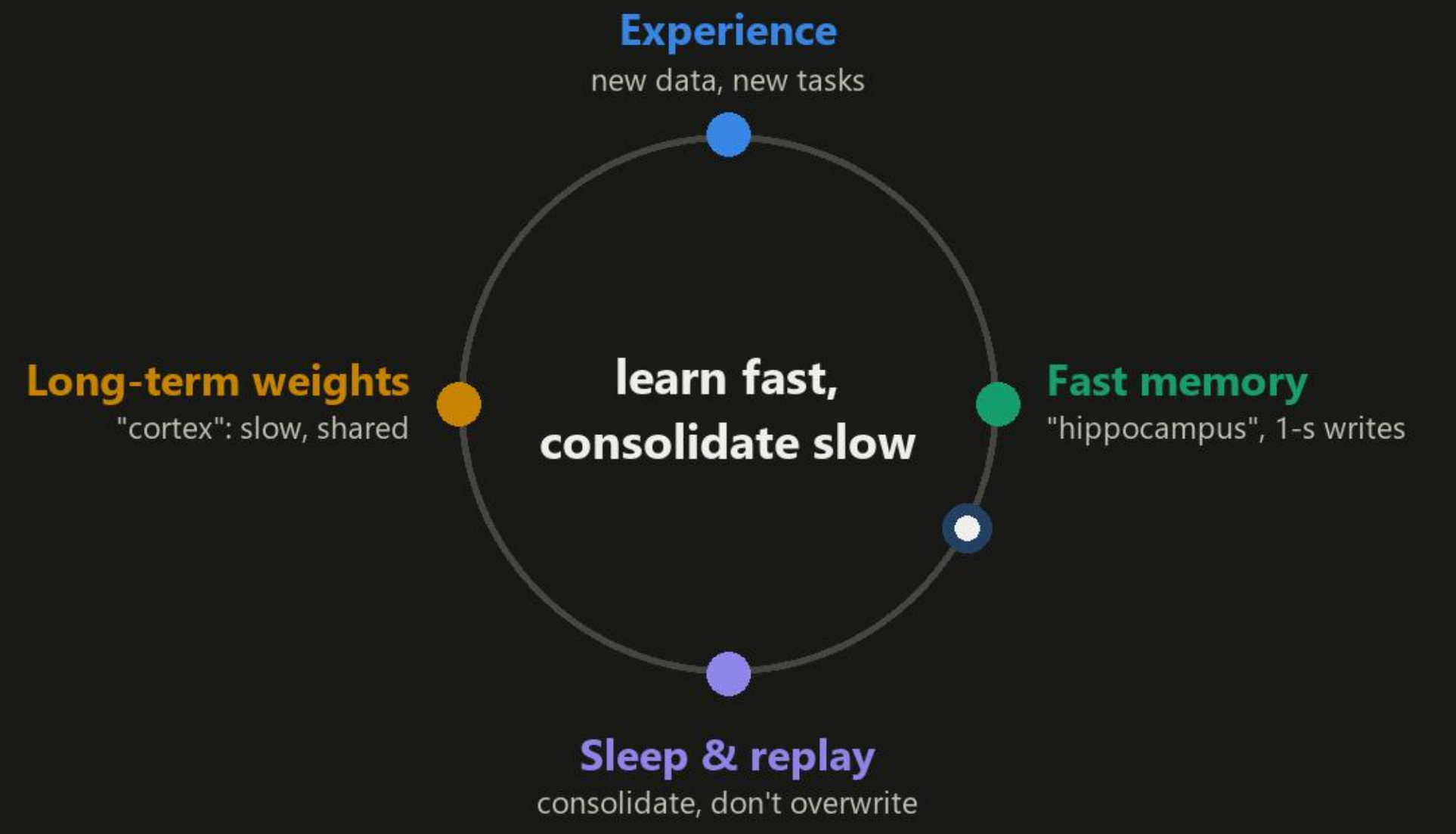



I spent two months testing ways around that.

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# My idea: learn like a brain

Fast memory for today. Slow weights for life. Sleep in between.




Inspired by complementary learning systems theory (hippocampus + cortex)

# What I built & tested

20+ experiments · 1 GPU (RTX 4080 SUPER) · ~$5 of cloud

## Memory

Fact slots on a frozen transformer · least-squares memory · sleep replay

## Continual learning

Re-tested Dohare et al. (Nature 2024) on loss of plasticity · 13 claims

## Reasoning from scratch

Trained GPTs 5M–124M · length extrapolation · composition · proofs

## Agents & tools

LoRA-tuned Qwen3-0.6B tool agent · Qwen2.5-14B memory gate · ARC-AGI-3

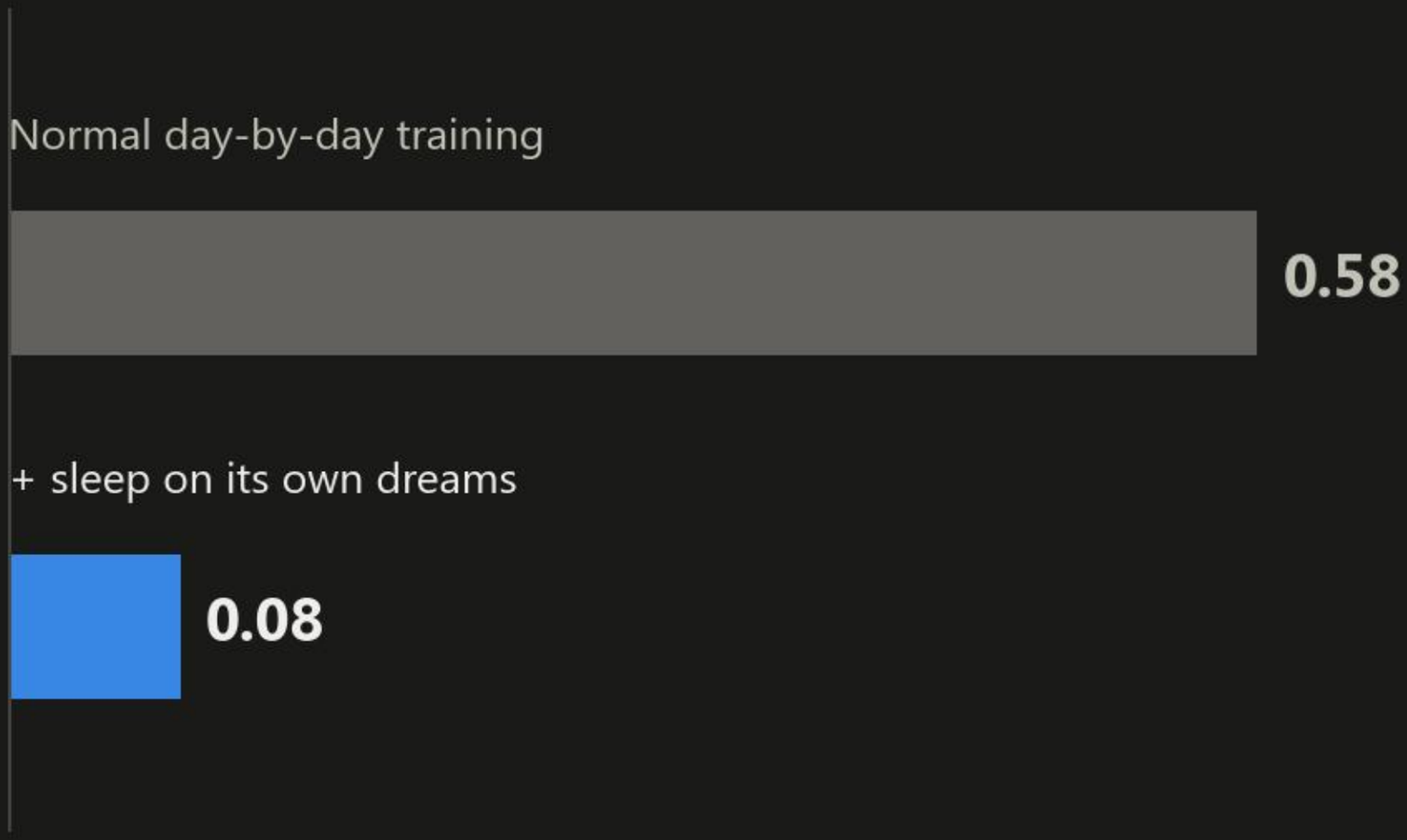
## Brain-inspired wiring

Real fruit-fly connectome vs RNN/GRU · DreamCoder-style program learning

Every run: pass/fail bar written first · controls · multiple seeds

# Sleep cut forgetting 7×

A small GPT learns day by day. Between days it 'sleeps': rehearses and replays its own generated text.

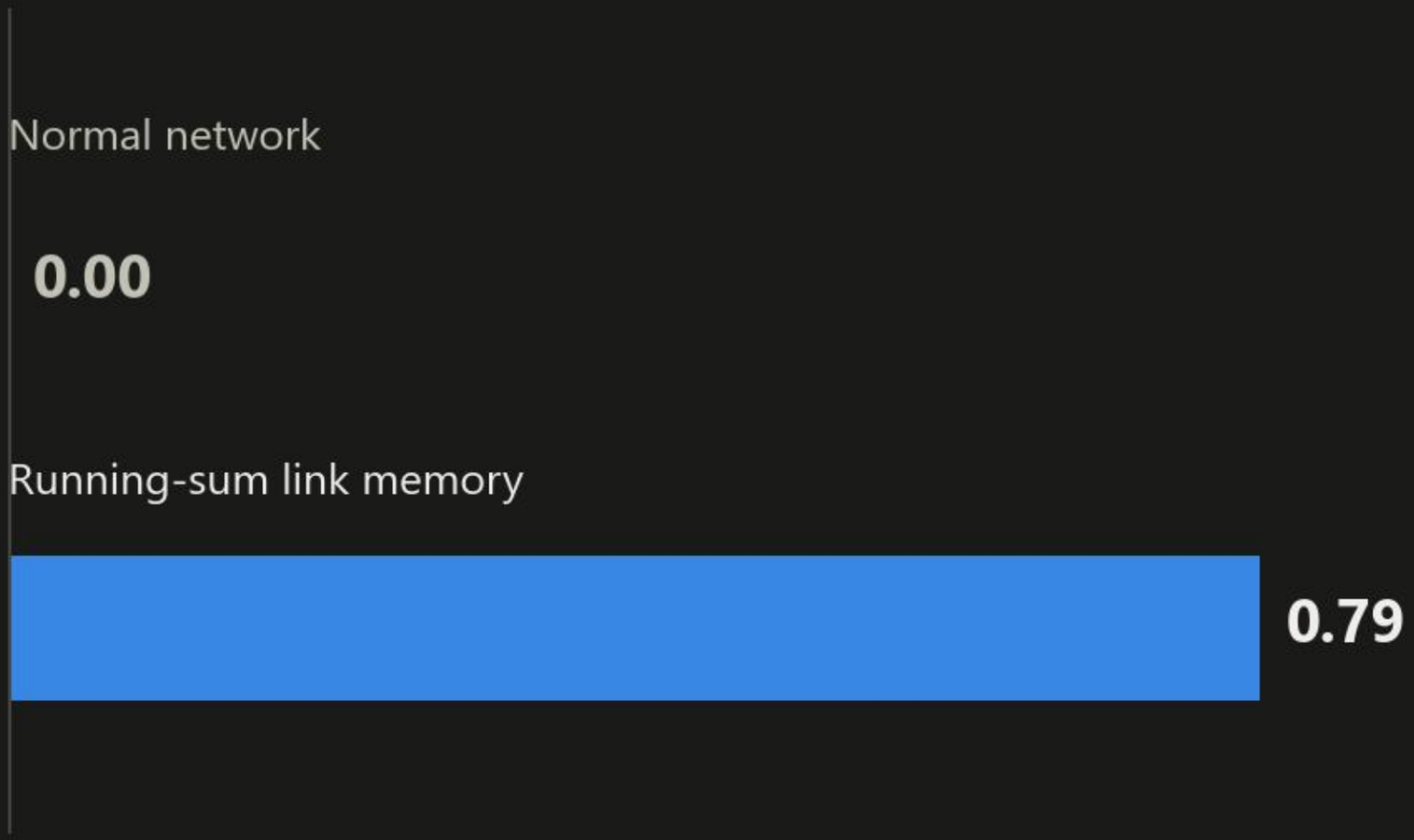



Forgetting in bits per byte, lower is better. Much less forgetting, but not zero.

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# Remembering day 1 after 20 days

CIFAR-100 learned as a 20-day stream, frozen image encoder. Accuracy on day-1 classes at the end.

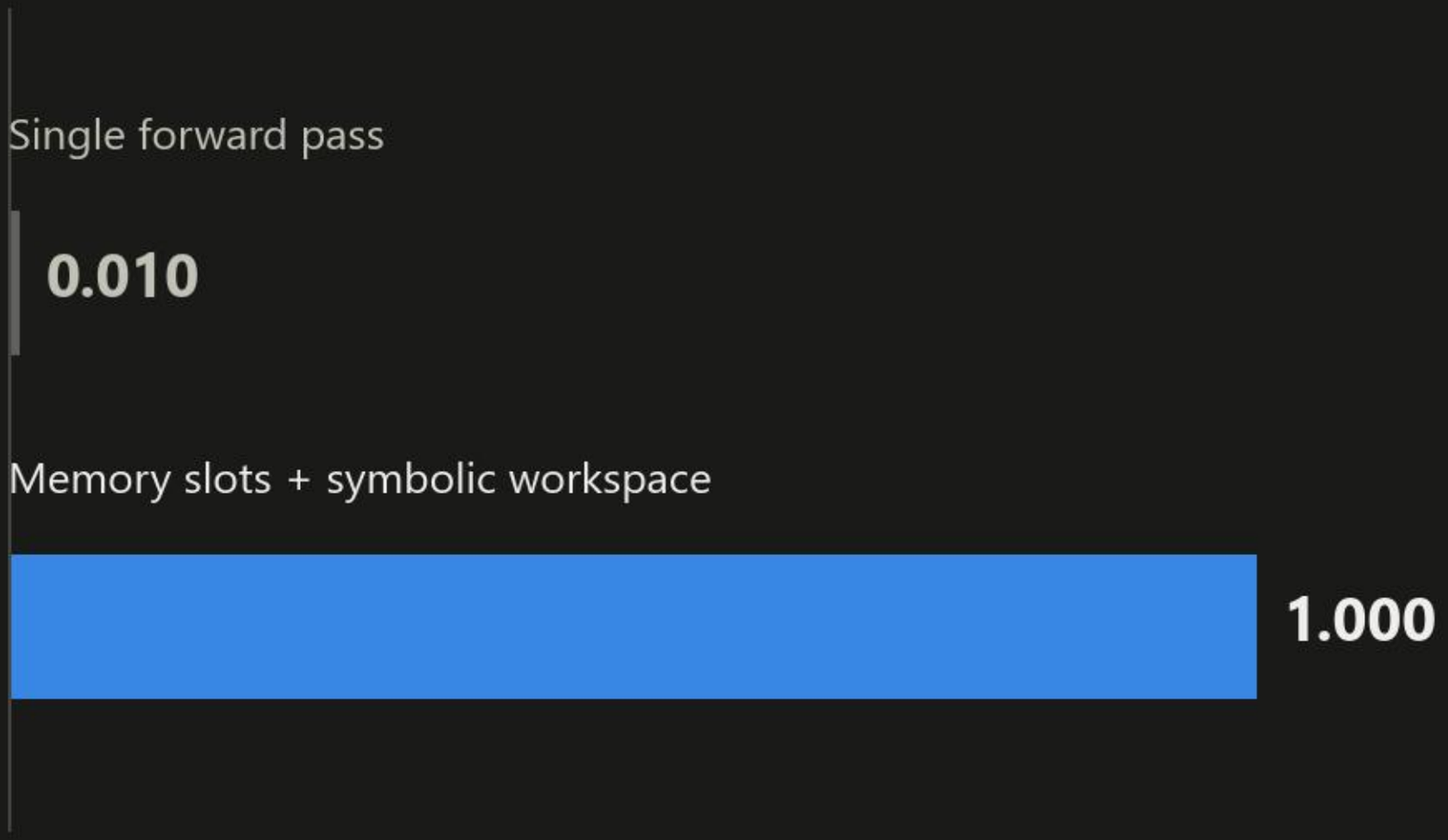



The link memory also matched learning everything at once on all classes (0.60 vs 0.60).

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# Memory slots on a frozen transformer

Write a new fact in ~25 steps (~1 s) without touching the backbone, then answer 2-hop questions.

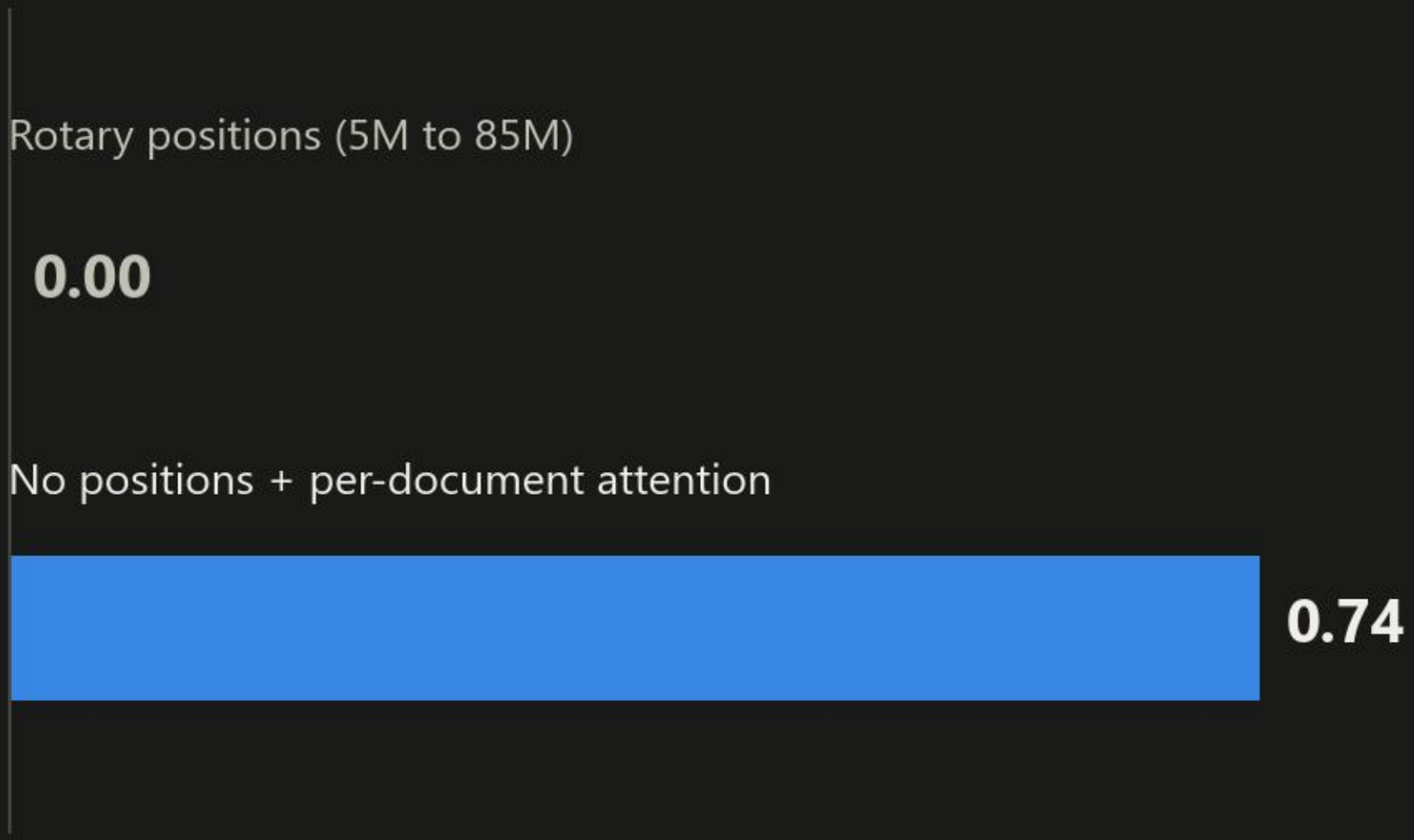



2-hop accuracy. 2,000 earlier facts stayed at 1.000, zero drift over 100 writes. Synthetic, closed vocabulary: unseen words still fail.

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# Counting past what it was trained on

Small GPTs trained from scratch in a mixed prose + reasoning stream.
Test: 6-digit addition, longer than training.

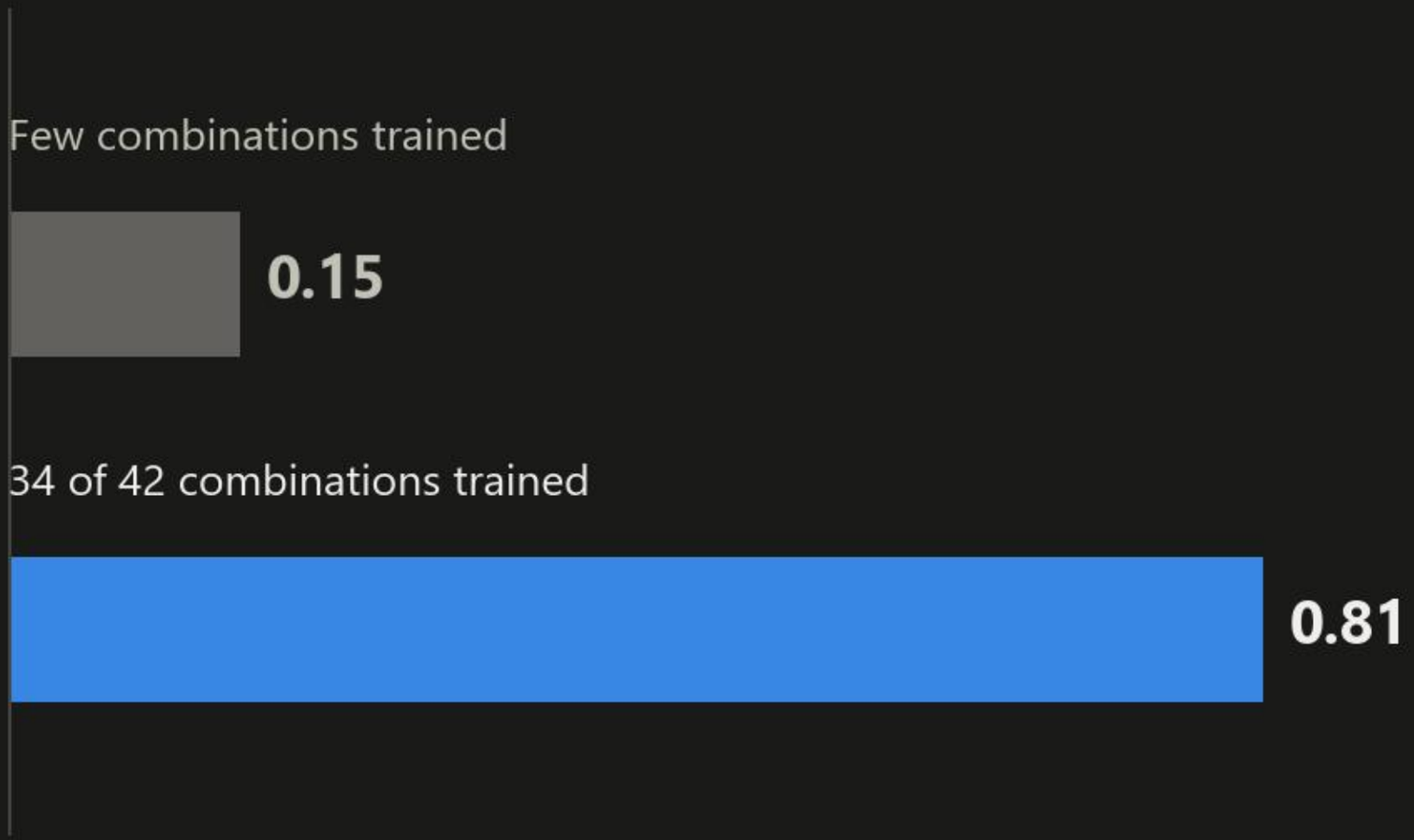



Scaling 5M → 85M left rotary at exactly 0.00. Removing positions and masking attention per document fixed it.

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# Composition is a data problem

Can a 5M model chain two skills it never saw chained, when asked in English?

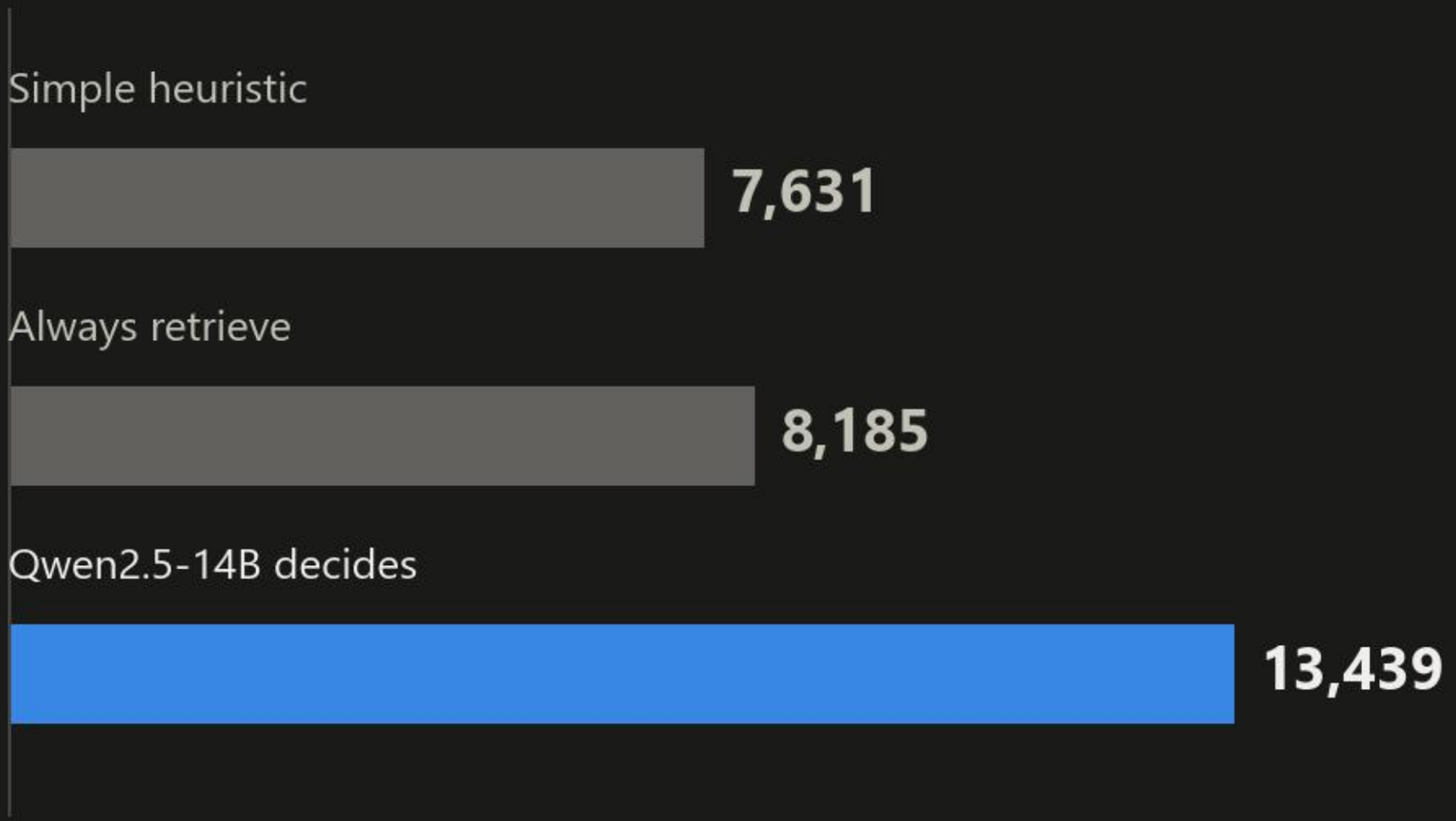



Accuracy on held-out pairs (low-diversity run: 0–15 of 100). More model size didn't help. More variety did.

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# A 14B model as memory gate

Qwen2.5-14B (local) as a retrieve-or-act gate vs simple policies, same tasks.




Total tokens, lower is better. The model gate also skipped two retrievals it needed. A negative result, logged as one.

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# Re-testing a Nature paper myself

13 claims from Dohare et al., "Loss of plasticity in deep continual learning" (Nature 2024) and a Rich Sutton interview. Permuted MNIST, 3 seeds, falsifier written before each run.




## Held up, e.g.

- Plain backprop loses plasticity
- Adam is worse
- L2 helps but doesn't eliminate it (−2.4 to −3.0 pts)
- Replay reduces forgetting

## Did not hold in my setup, e.g.

- Shrink-and-perturb nearly eliminates the loss
- "Forgetting is totally curable" with continual backprop
- The three correlates all move as described

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# Results that held up

Small models, careful tests, numbers from my own runs

**7×**

**less forgetting**

GPT sleeps on its own dreams: 0.58 → 0.08 bits/byte

**2,000**

**old facts kept at 100%**

while new facts are written in ~1 s each

**0.74**

**length extrapolation**

rotary positions: 0.00 → no positions + per-doc attention

**0.81**

**unseen compositions**

up from ≤ 0.15 by adding data diversity, not size

Failures are logged too: real fly wiring lost to a GRU; a 14B model deciding when to retrieve used 64–76% more tokens than a simple rule.

# What I learned

**01** Simple baselines beat clever mechanisms more often than you'd think. Better to learn that in a day.

**02** Generalization came from data diversity and outside verification, not model size or self-checking.

**03** Dense shared weights forget. Replay, sleep and regularization only partly fix it.

NEXT

**Architectures where new learning doesn't overwrite old knowledge, running on a local device.**

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